Exercise Sheet 1

Even though this PVK is divided into sections and some content has not yet been covered, the entire material covered in the semester and in the lecture may and should be explicitly used in the series.

Exercise 1
Show that a sequence is convergent if and only if it is a Cauchy sequence.

Exercise 2
Compute the following limits.
Hint: You can use Bernoulli-L'Hospital and Taylor-Approximations.

$$\lim_{x\to 1} \frac{\sqrt{x}-1}{x-1}$$

$$\lim_{x\to 0} \frac{\tan(x)-\sin(x)}{x^3}$$

$$\lim_{x\to 0^+} \frac{e^{-\frac{1}{x}}}{x}$$

Exercise 3 (FS2024, A. Figalli)
Calculate the following Limit.

$$\lim_{x\to 0} \frac{x+\cos(x)\sin(3x)}{\sin^3(x)+3\sin(x)-x}.$$

Exercise 4 (HS2022, G. Felder)
Calculate the following Limit.

$$\lim_{x\to \pi} \frac{\sin^2(x)}{\cos(x)+1}.$$

Exercise 5
Let $D \subset \mathbb{R}$ be a set, let $f : D \to \mathbb{R}$ be a function, and let $\bar{x} \in D$. Show that the function f is continuous at $\bar{x}$ if and only if, for all sequences $(x_n)_n \subset D$ converging to $\bar{x}$, $f((x_n)_n)$ converges to $f(\bar{x})$.

Exercise 6
Show that a Lipschitz continuous function f is also uniformly continuous. Give an example of an uniformly continuous function that is not Lipschitz continuous.

Exercise 7
Briefly explain why a uniformly continuous function is always continuous, but a continuous function is not always uniformly continuous.
Note: This does not need to be very rigoros, but a recap of the definitions.

Exercise 8 (HS2022, G. Felder) (a) Let $(f_n)_n$ be a sequence of continuous functions on $D \subset \mathbb{R}$ that converges uniformly to a function f. Show that f is also continuous.

(b) Give an example of a sequence of continuous functions that converge pointwise to a non-continuous function.

Exercise 9
We look at the sequence of functions f_n for $n \geq 1$ defined as follows

$$f_n(x) := \frac{x}{1 + nx^2}.$$

(A) Are the functions f_n uniformly continuous? Justify your answer!

(B) Determine the pointwise limit of this sequence of functions.

(C) Do the functions f_n converge uniformly? Justify your answer!

Exercise 10
Let $f : [a, b] \to \mathbb{R}$ be a continuous function with $f(a) \leq f(b)$. Show that for every real number c with $f(a) \leq c \leq f(b)$, there exists a $\bar{x} \in [a, b]$ such that $f(\bar{x}) = c$.

Exercise 11
Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Prove that f is also uniformly continuous on $[a, b]$.